Containers for HPC

Getting Started with Apptainer

HPC Research Computing Workshop · 60 min · 20 min slides + 40 min hands-on

What Problem Are We Solving?

> "It works on my laptop"
> The classic research tragedy — your analysis runs fine locally but crashes on the cluster or a collaborator's machine.

> Irreproducibility
> You can't re-run your own analysis 6 months later because the environment has changed. Reviewers can't reproduce your results.

What Is a Container?

📦 The Shipping Container Analogy

- A standardized box that works on any ship, truck, or dock
- Your container works on your laptop, the cluster, and your collaborator's machine

Container vs. Virtual Machine

Feature	Container	VM

Docker vs. Apptainer: Why Not Just Use Docker?

Feature	Docker	Apptainer
Requires root/sudo	✅ Yes	❌ No

Runs as your user	❌ No	✅ Yes

How Apptainer Works on HPC

1 · Definition File

2 · Build Once

3 · One File

4 · Run Everywhere

Key Apptainer Commands

90% of your day-to-day usage fits in four commands

`apptainer pull` — Download a pre-built image from Docker Hub. Millions available. No build required.

Definition Files: Your Reproducibility Recipe

▒▒▒▒ Why use `.def` files?

```
# myenv.def
```

- 📄 Version control — check into Git alongside your code
- 🔄 Rebuild anytime — anyone can recreate the exact environment

```
Bootstrap: docker        # start from Docker Hub
```


Real Research Use Cases

🧬 Bioinformatics

🐍 Python / R Science

- Pin exact GATK, BWA, Samtools versions
- Share complete pipeline with your paper

- No more conda environment conflicts
- Isolate per-project dependencies

Typical HPC Workflow

```
| 1. BUILD |──▶| 2. STORE |──▶| 3. TEST |──▶| 4. RUN |
```

Tips & Gotchas

🔧 Building — needs root or a workaround

🖥️ GPU Jobs — the `--nv` flag

```
# Option 1: fakeroot (most clusters)
```

```
# Pass through NVIDIA GPU from host
```

HANDS-ON DEMOS

Demo 1 — Pull & Explore a Pre-built Image

⏱ 8 min · "Zero to running in 2 minutes"

Step 1 — Pull Python 3.11 from Docker Hub

Demo 2 — Build a Simple Science Image

⏱ 10 min · "Your first definition file"

Demo 3 — Bioinformatics Pipeline Container

⏱ 10 min · "Real-world research example — FastQC + MultiQC"

Demo 4 — SLURM Job Script Integration

⏱ 7 min · "Submitting container jobs to the scheduler"

Demo 5 — GPU Container: PyTorch + CUDA

⏱ 5 min · "NVIDIA GPU passthrough — the --nv flag"

Key Takeaways & Resources

Key Takeaways

- 📦 One `.sif` file = your entire software environment, frozen in time Not a requirements.txt — the `.sif` IS the environment, ready to run

- 👤 Runs as YOU — same UID inside and outside. No root, no risk. This is why sysadmins allow it — it can't escalate privileges
- 🐳 The entire Docker Hub ecosystem works via `docker://` prefix Millions of images: Python, R, CUDA, Bioconductor, GATK, and more
- 📄 `.def` files in Git = reproducible, peer-reviewable science Your environment is code — version control it like code

Resources

Resource	URL	Notes
📖 Apptainer docs	apptainer.org/docs	Official — installation, CLI reference